

The Near-Critical Sample Tradeoff for Uniform-Norm Projections

Endpoint resolution and quantitative bounds for arXiv:2401.02220

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Status. Candidate partial result, likely valid. The case of one extra sample (equivalently $\alpha = 0$) is completely resolved up to absolute constants. For $0 < \alpha < 1$ the packet gives new explicit brackets, but the full asymptotic tradeoff remains open.

1 Source problem

Krieg, Pozharska, M. Ullrich, and T. Ullrich prove that every n -dimensional subspace of $B(D)$ admits a sampling projection using $2n$ function values and having norm $O(\sqrt{n})$ [1]. At the other endpoint, n samples require norm of order n in the worst case. They ask for the smallest universal norm when only $n + n^\alpha$ values are allowed, $0 \leq \alpha < 1$. The exact source passage is reproduced below.

To make “smallest universal norm” precise, let

$$\Pi(n, m) = \sup_{D, V} \inf_P \|P\|_{B(D) \rightarrow B(D)},$$

where the supremum is over all sets D and all complex n -dimensional subspaces $V \subset B(D)$, and the infimum is over sampling projections

$$Pf = \sum_{i=1}^r f(x_i) \varphi_i, \quad r \leq m, \quad x_i \in D, \quad \varphi_i \in V,$$

onto V . Repeated sample points may be merged, so they never improve the infimum.

2 Result

Theorem 1 (Near-critical sampling bracket). *For $n \geq 2$ and $1 \leq k \leq n$,*

$$\Pi(n, n+k) \geq \max \left\{ \frac{n}{\sqrt{n+k}}, \left\lfloor \frac{n}{k+1} \right\rfloor \right\}. \quad (1)$$

For every such k one has $\Pi(n, n+k) \leq n+1$. If $k \geq 2$, then also

$$\Pi(n, n+k) \leq \sqrt{n+2} \frac{\sqrt{n+k} + \sqrt{n+1}}{\sqrt{n+k} - \sqrt{n+1}} \quad (2)$$

$$\leq \frac{4(n+k)\sqrt{n+2}}{k-1}. \quad (3)$$

Consequently the better of $n+1$ and (2) may always be used.

The main goal in this paper is to show that also the projection in the Kadets-Snobar theorem can be given by a sampling projection and thus improving on Novak's result by a factor of order $\sqrt{n}$. However, in contrast to Novak, we require a slight *oversampling*, i.e., we need $m \geq cn$, $c > 1$, function evaluations for subspaces V_n of dimension n .

Theorem 1. *There is an absolute constant $C > 0$ such that the following holds. Let D be a set and V_n be an n -dimensional subspace of $B(D)$. Then there are $2n$ points $x_1, \dots, x_{2n} \in D$ and functions $\varphi_1, \dots, \varphi_{2n} \in V_n$ such that $P: B(D) \rightarrow V_n$ with $Pf = \sum_{i=1}^{2n} f(x_i) \varphi_i$ is a projection onto V_n with $\|P\| \leq C\sqrt{n}$.*

Theorem 1 is sharp in the following sense:

- Using $m = \mathcal{O}(n)$ samples, the norm bound $C\sqrt{n}$ cannot be replaced with a lower-order term, see Section 5.3.
- If we want to use $m = n$ samples, the norm bound $C\sqrt{n}$ has to be replaced by a linear term in n , see [33, Rem. 1.2.3].

The oversampling factor 2 can be replaced by any constant $c > 1 + 1/n$, see Remark 7. It is an interesting open problem, what the smallest possible norm is when using only $n + n^\alpha$ samples with $0 \leq \alpha < 1$.

Theorem 1 is based on a specific type of *discretization* of the uniform norm, which might be of independent interest. In particular, Theorem 1 will be deduced from the following and Lemma 14, see Section 4.

Theorem 2. *Let D be a set and V_n be an n -dimensional subspace of $B(D)$. Then there are points $x_1, \dots, x_{2n} \in D$ such that, for all $f \in V_n$,*

Figure 1: The open problem on page 2 of arXiv:2401.02220v2.

Corollary 2 (Power oversampling). *Let $k_n = \lceil n^\alpha \rceil$. For $\alpha = 0$,*

$$\Pi(n, n+1) = \Theta(n).$$

Thus the first previously open point of the tradeoff curve is fully resolved. For fixed $0 < \alpha < 1$,

$$n^{\max\{1/2, 1-\alpha\}} \lesssim \Pi(n, n+k_n) \lesssim n^{\min\{1, 3/2-\alpha\}}. \quad (4)$$

In particular, the upper bound becomes nontrivial already for $\alpha > 1/2$. The unweighted estimate stated in Remark 7 of the source paper becomes nontrivial only for $\alpha > 2/3$.

3 Proof intuition

The two lower bounds detect different obstructions. Linear functions on the Euclidean sphere force every m -sample reconstruction of the identity to have norm at least $n/\sqrt{m}$; this follows just by taking a trace and then averaging over random phases. To see the instability at the critical sample count, use the zero-sum hyperplane

$$H_q = \{z \in \mathbb{C}^{q+1} : z_0 + \dots + z_q = 0\}.$$

Sampling exactly q of its $q + 1$ coordinates forces the omitted coordinate to be minus the sum of the sampled values, and hence forces projection norm q . Taking $k + 1$ disjoint blocks guarantees that one block receives no extra sample, which gives the second term in (1).

The upper bound uses positivity in a different way. Adjoin the constant function to V . A Kiefer–Wolfowitz design gives a probability measure for which the uniform norm is bounded by $\sqrt{n + 1}$ times the L^2 norm. Weighted BSS sparsification reduces this measure to $n + k$ atoms while losing a factor of order n/k in the L^2 comparison. The adjoined constant controls the total sparse weight. Weighted least squares is then a sampling projection with the same order of norm. This last constant-function device is essential: without it, the sparse weights need not have controlled total mass.

4 Lower bounds

Lemma 3 (Euclidean trace obstruction). *For every $m \geq n$, $\Pi(n, m) \geq n/\sqrt{m}$.*

Proof. Let D be the Euclidean unit sphere of $\mathbb{C}^n$, and identify $\mathbb{C}^n$ with the space V of linear functions $f_z(x) = \langle z, x \rangle$ on D . This is an isometry from ℓ_2^n into $B(D)$.

Suppose a sampling projection uses distinct points $x_1, \dots, x_r$, where $r \leq m$. Write its output functions as $f_{a_1}, \dots, f_{a_r}$. Since the operator is the identity on V ,

$$z = \sum_{i=1}^r \langle z, x_i \rangle a_i \quad (z \in \mathbb{C}^n). \quad (5)$$

Taking traces and then absolute values gives

$$n \leq \sum_{i=1}^r \|a_i\|_2 \leq \sqrt{r} \left(\sum_{i=1}^r \|a_i\|_2^2 \right)^{1/2}.$$

For independent uniform phases ε_i on the unit circle,

$$\mathbb{E} \left\| \sum_i \varepsilon_i a_i \right\|_2^2 = \sum_i \|a_i\|_2^2.$$

Because an arbitrary bounded function on D may take the prescribed values ε_i at the distinct sample points, the square root on the right is at most the projection norm. Hence $n \leq \sqrt{r} \|P\| \leq \sqrt{m} \|P\|$. $\square$

Lemma 4 (Block-hyperplane obstruction). *For $1 \leq k \leq n$,*

$$\Pi(n, n + k) \geq \left\lfloor \frac{n}{k + 1} \right\rfloor.$$

Proof. Put $b = k + 1$ and $q = \lfloor n/b \rfloor$. On a set of $q + 1$ points let

$$H_q = \{z \in \mathbb{C}^{q+1} : \sum_{i=0}^q z_i = 0\},$$

with its inherited uniform norm. Form the ℓ_∞ -direct sum of b disjoint copies of H_q and add $n - bq$ one-dimensional coordinate blocks. The resulting function space $V \subset B(D)$ has dimension n .

Any sampling projection onto V must use at least q distinct points in each H_q block and at least one in each one-dimensional block: otherwise the restriction of the evaluation map to that summand is not injective. These compulsory samples account for n points. There are only $k = b - 1$ extra points, so at least one H_q block is sampled at exactly q of its $q + 1$ coordinates.

Restrict inputs to that block and take the corresponding output component. This gives a q -sample projection from ℓ_∞^{q+1} onto H_q . The q sampled-coordinate functionals restrict to an isomorphism on H_q , so its left inverse is unique: it copies the sampled values and puts minus their sum in the omitted coordinate. Its norm is exactly q . Therefore the original projection also has norm at least q . $\square$

Lemmas 3 and 4 prove (1). Notice that the second construction is what survives one extra sample: a single zero-sum block becomes harmless after sampling all its coordinates, but $k+1$ blocks ensure that k extra samples cannot repair all of them.

5 Upper bound

We use the following finite-frame sparsification consequence of Bartel–Schäfer–Ullrich [2]. It is their preconditioned complex BSS estimate, specialized to a finite-dimensional subspace.

Lemma 5 (Weighted near-minimal discretization). *Let W be a d -dimensional subspace of $L^2(\rho)$, where ρ is a finitely supported probability measure. For every $c > 1$ there are at most $\lceil cd \rceil$ atoms x_j and weights $\lambda_j \geq 0$ such that*

$$\|w\|_{L^2(\rho)}^2 \leq \sum_j \lambda_j |w(x_j)|^2 \leq \gamma_c \|w\|_{L^2(\rho)}^2, \quad \gamma_c = \left(\frac{\sqrt{c} + 1}{\sqrt{c} - 1} \right)^2. \quad (6)$$

Proof of the upper bounds in Theorem 1. The $n+1$ bound is the sampling version of Auerbach’s lemma recalled in Remark 4 of [1]; extra samples may simply be ignored.

Assume $m = n + k \geq n + 2$. Set $W = \text{span}(V \cup \{1\})$ and $d = \dim W \leq n + 1$. Proposition 8 of [1], applied with its parameter $\varepsilon = 1$, gives a finitely supported probability measure ρ such that

$$\|w\|_\infty \leq \sqrt{d+1} \|w\|_{L^2(\rho)} \quad (w \in W). \quad (7)$$

Apply Lemma 5 with $c = m/d > 1$. Let $\|f\|_\nu^2 = \sum_j \lambda_j |f(x_j)|^2$. Since $1 \in W$, the upper half of (6) gives

$$\nu(D) = \sum_j \lambda_j = \|1\|_\nu^2 \leq \gamma_c. \quad (8)$$

Let $A_\nu f$ be the weighted least-squares minimizer over V . The lower half of (6) makes this minimizer unique; it is linear, depends on at most m values of f , and is the identity on V . Thus it is a sampling projection. Orthogonal projection in the ν inner product, (7), (6), and (8) give

$$\begin{aligned} \|A_\nu f\|_\infty &\leq \sqrt{d+1} \|A_\nu f\|_{L^2(\rho)} \leq \sqrt{d+1} \|A_\nu f\|_\nu \\ &\leq \sqrt{d+1} \|f\|_\nu \leq \sqrt{(d+1)\gamma_c} \|f\|_\infty. \end{aligned}$$

Therefore

$$\|A_\nu\| \leq \sqrt{d+1} \frac{\sqrt{m} + \sqrt{d}}{\sqrt{m} - \sqrt{d}} \leq \sqrt{n+2} \frac{\sqrt{m} + \sqrt{n+1}}{\sqrt{m} - \sqrt{n+1}},$$

which is (2). Finally,

$$\frac{\sqrt{m} + \sqrt{d}}{\sqrt{m} - \sqrt{d}} = \frac{(\sqrt{m} + \sqrt{d})^2}{m - d} \leq \frac{4m}{m - d} \leq \frac{4(n+k)}{k-1},$$

proving (3). $\square$

Corollary 2 follows immediately by putting $k = \lceil n^\alpha \rceil$ in Theorem 1.

6 Upgrade attempts, verification, and remaining gap

The proof was pushed beyond the first partial observation in four ways. First, the unweighted sparsification used in the source was replaced by the weighted preconditioned BSS estimate, and the constant function was adjoined to control total mass. Second, two independent lower witnesses were made explicit rather than importing the paper’s sharpness assertion. Third, tensor products of zero-sum hyperplanes were tested as possible stronger obstructions. Exact small linear programs give best norms 3, 2 for $H_2 \otimes H_2$ with 4, 5 samples and 4, 3, 2 for $H_2 \otimes H_3$ with 6, 7, 8 samples; the rapid improvement rules out the naive tensor route. Fourth, an interpolation-plus-rank-one-correction decomposition was examined. Although each correction can be chosen to annihilate V and hence preserve the projection property, no selection principle controls the uniform operator norm of a partial correction sum; the necessary cancellations are global in the output rows.

The script `code/verify_finite_witnesses.py` checks the unique q -sample projection onto H_q for $1 \leq q \leq 12$, verifies its norm is q , and checks the block-count formulas and scalar bounds in the theorem. These computations are consistency checks and are not used in the proof.

For $0 < \alpha < 1$, the exponent gap in (4) remains. Closing it appears to require either a sampling projection construction that uses cancellations unavailable to positive weighted least squares, or a single family of subspaces whose instability persists quantitatively under many extra coordinates. The block construction only charges one extra sample to repair one hard block, and weighted spectral sparsification intrinsically pays the $(c - 1)^{-1}$ loss.

7 Novelty assessment

The exact title, arXiv identifier, open-problem phrase, “ $n + k$ sampling projection”, “near-critical oversampling”, and related frame- sparsification terms were searched through 11 August 2026. The October 2025 version of the source still poses the problem. A March 2026 survey by Dai, Kosov, and Temlyakov [3] gives a weighted norming-set estimate of order $\sqrt{n}/(c - 1)$ by adjoining the constant function, but does not state the sampling-projection tradeoff, the block lower bound, or the one-extra- sample resolution. Thus the upper argument should be viewed as a short projection consequence of known ingredients, not as a wholly new sparsification theorem. No source located in the bounded search contains the block-hyperplane lower bound or the conclusion $\Pi(n, n + 1) = \Theta(n)$. Novelty confidence for those additions is moderate, not definitive.

Human-review recommendation. Review as a likely-valid partial result with a fully resolved endpoint. The highest-value checks are the uniqueness argument on an exactly sampled zero-sum block, the count forcing one such block, and the use of the constant function in (8). The packet makes no claim to resolve the interior range $0 < \alpha < 1$.

References

- [1] D. Krieg, K. Pozharska, M. Ullrich, and T. Ullrich, *Sampling projections in the uniform norm*, J. Math. Anal. Appl. **553** (2026), Paper 129873; arXiv:2401.02220.
- [2] F. Bartel, M. Schäfer, and T. Ullrich, *Constructive subsampling of finite frames with applications in optimal function recovery*, Appl. Comput. Harmon. Anal. **65** (2023), 209–248; arXiv:2202.12625.
- [3] F. Dai, E. Kosov, and V. N. Temlyakov, *A survey of sampling discretization of integral and uniform norms*, arXiv:2603.02459 (2026).